METRIC

MIL-M-24614A(SH)
12 February 1993
SUPERSEDING
DOD-M-24614(SH)
21 January 1983

MILITARY SPECIFICATION

METERING SYSTEM, STORAGE BATTERY CURRENT AND AMPERE-HOUR, DIGITAL (METRIC, NONMERCURY TYPE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a digital current and ampere-hour meter (AHM) system for a submarine or surface ship storage battery circuit.

1.2 Classification. Metering systems shall be of the following classes and types as specified (see 6.2)

1.2.1 Class. Classes shall be determined by the application and shall be as follows:

Class I - Submarine application, TRIDENT, SSN 688 and SSN 637 - with ampere display suppressed

Class II - Surface ship application - with ampere display operating

Class III- Submarine application (SEAWOLF) - With ampere display operating and combined ampere-hour display and totalizing ampere-hour displays for integration into the Electric Plant Control Panel (EPCP), no enclosure

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, SEA 05Q42, Commander Naval Sea Systems Command, 2531 National Center Bldg 3, Washington, DC 20362-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 6625

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1.2.2 Type. Type shall be determined by the shunt rating values. Types shall be as follows:

Type 0 - 2,000 amp/100 mV shunt/50 μohm
Type I - 2,500 amp/100 mV shunt/40 μohm
Type II - 3,000 amp/100 mV shunt/33.33 μohm
Type III - 4,000 amp/100 mV shunt/25 μohm
Type IV - 5,000 amp/100 mV shunt/20 μohm
Type V - 8,000 amp/100 mV shunt/12.5 μohm
Type VI - 10,000 amp/100 mV shunt/10 μohm

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein (see 6.2). Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-S-901	Shock Tests, HI (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-915	Cable and Cord, Electrical, For Shipboard Use, General Specification for
MIL-E-917	Electric Power Equipment, Basic Requirements for (Naval Shipboard Use)
MIL-E-2036	Enclosures for Electric and Electronic Equipment
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment
MIL-P-15024/5	Plates, Identification
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts); Packaging of

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MIL-L-19140	Lumber and Plywood, Fire Retardant Treated
MIL-S-19622	Stuffing Tubes, Nylon; and Packing Assemblies; General Specification for
MIL-C-21097	Connectors, Electrical, Printed Wiring Board, General Purpose, General Specification for
MIL-E-24142	Enclosures for Electrical Fittings and Fixtures, General Specification for
MIL-E-24142/6	Enclosure, Submersible, (15 foot) Size 8 by 10 through 14 by 26
MIL-C-24643	Cable and Cord, Electrical, Low Smoke, for Shipboard use, General Specification for
MIL-C-24643/33	Cable, Electrical, Type LS2SWU including Variation LS2SWUA)
MIL-T-55164	Terminal Lugs Solder Type, Styles TBLD and TBLS

STANDARDS

MILITARY

MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-781	Reliability Testing for Engineering Development, Qualification and Production
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1399, Section 300	Interface Standard for Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-2036	General Requirements for Electronic Equipment Specifications

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATION NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-00-312-8010 Maintainability Design Criteria Handbook for Designers

(Application for copies should be addressed to the Naval Publications and Forms Center, (A: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

200-75 Electrical and Electronic Parts and Equipments IEEE Standard Reference Designation for (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.6) in accordance with 4.3.

3.2 Materials. Materials used in the construction of the metering systems shall be as specified herein.

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3.2.1 Prohibited materials.

3.2.1.1 Mercury. The use of the element mercury in any component of the metering systems and for any manufacturing process is prohibited.

3.2.1.2 Thermionic devices. The use of thermionic devices such as electron tubes is prohibited.

3.2.2 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 System assemblies. The current measuring and display subsystems shall consist of the following functional module assemblies.

a. **Ammeter function.** The ammeter shall be a digital module of the specified class (see 1.2.1) and type (see 1.2.2). This functional module shall sense the voltage across the appropriate shunt and convert this signal into digital information. This information will be used to provide a direct reading display of current when required (see 1.2.1) and for the purpose of integration by the AHMs described in 3.3(b). The display shall consist of a minus sign (indicating discharge), no sign (indicating charge), and five decimal digits. A display of "OL" shall indicate an overload condition. Currents as large as ± 9,999.8 amperes with a minimum resolution of one ampere for currents equal to or greater than 1,000 amperes and a minimum resolution of 0.1 ampere for currents less than 1,000 amperes shall be capable of being displayed. The leading zeros in the first three digits shall be suppressed. Class I meters shall provide only overload and charge/discharge indication.

b. **Ampere-hour meters function.** Two, digital AHM displays shall be provided. One of these shall digitally display the integral of the positive-polarity output and the other shall display the integral of the negative-polarity output of the ammeter function. Except for the front-panel legends, "charge" or "discharge", the two AHMs shall be identical and interchangeable. A momentary pushbutton switch shall be provided for each display on the front panel of the system which will permit the operator to clear the accumulated value to zero. The display shall be a seven-digit decimal with the decimal point to the left of the least significant digit. The display shall suppress leading zeros, in the first five digits and shall provide indication of overflow beyond 999,999.9 AMP HRs. The ampere hour module shall be capable of retaining the accumulated AMP HR reading for a period of at least one week in the event of a power line failure. There shall be no loss of stored data upon power interruption or restoration.

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c. **Totalizing ampere-hour meter function.** Two digital totalizing AMP HR displays shall be provided. One shall display charge AMP HRs and the other discharge AMP HRs. The totalizing display be in the form of seven decimal digits to a maximum value of 9,999,999 AMP HRs. Totalizing displays shall be normally suppressed and a pushbutton switch shall be provided to permit the operator to enable the displays momentarily when desired. Except for the front-panel legends, "charge" or "discharge", the two totalizing displays shall be identical and interchangeable. Leading zeros, except for integer AMP HRs shall be suppressed in the display, and an indication of overflow beyond 9,999,999 AMP HRs shall be provided. Each totalizing display shall be capable of being manually cleared to zero after the front face of the indicator is removed.

d. **Class III displays.** There shall be two dual purpose ampere-hour displays: one for charge ampere-hours and one for discharge ampere-hours. There shall be one operating ammeter display. The ampere-hour meter function and the totalizing ampere-hour meter function shall be combined in a single case with separate push buttons for displaying total ampere-hours and resetting the ampere-hour meter.

3.4 Interconnections. Unless otherwise specified (see 6.2), cables shall not be furnished. When cables are furnished, the connections between the current shunt, the metering system, and the totalizing displays shall be made by shielded cables having a maximum resistance of 2.3 ohms per conductor per 100 meters at 25 degrees Celsius (°C) (type LS2SWU of MIL-C-24643 and MIL-C-24643/33). Maximum cable lengths will be 100 meters between the current shunt and the meter system.

3.4.1 Signal cable stuffing tube. For Classes I and II the signal cable stuffing tube shall conform to Military part number M19622/1-001 in accordance with MIL-S-19622 and MIL-S-19622/1.

3.4.2 Power cable stuffing tube. For Classes I and II the power cable stuffing tube shall conform to Military part number M19622/1-002 in accordance with MIL-S-19622 and MIL-S-19622/1.

3.4.3 Stuffing tubes. Stuffing tubes shall conform to MIL-S-19622.

3.5 Construction of assemblies.

3.5.1 General. The construction shall include, to the greatest extent possible, features that facilitate maintenance. All displays (or display modules) shall be removable for ease of repair or replacement. Printed wiring boards within the instruments shall be in accordance with requirement 17 of MIL-STD-454. The display digits shall be of the luminous type (either plasma discharge or light emitting diode (LED)), orange in color, at least 1 centimeter (cm) in height, and easily readable in normal room illumination (750 Lux) at the face of the display over an angular range of at least plus or minus 45 degrees. The display shall remain readable at a distance of 2 meters minimum. Socketed components shall be secure against becoming dislodged by shock or vibration. Circuit board contact fingers used for interconnection with edgeboard connectors shall be gold plated The front panel of the ammeter shall be as specified on figure 1 The ammeter displays shall be labeled. "AMPERES", the AHM displays shall be labeled "DISCHARGE AMP-HOURS" OR CHARGE AMP-HOURS

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as appropriate, and the totalizing AMH displays shall be labeled "TOTAL DISCHARGE AMP-HRS" or "TOTAL CHARGE AMP-HOURS," as appropriate. The pushbutton on the front panel shall be in accordance with figure 1 and be labeled: "RESET."

3.5.2 Terminal boards. Terminal boards shall be in accordance with MIL-T-55164.

3.5.3 Identification plates. The current measuring system shall be equipped with an identification plate in accordance with MIL-P-15024. The identification plates shall conform to the marking requirements of MIL-P-15024/5.

3.5.4 Enclosure. For Classes I and II the metering system shall be mounted in a Military part number M24142/6-001, size 8 by 10, enclosure conforming to MIL-E-24132/6 and MIL-E-24132 with a type "B" (flat) cover. The enclosure shall be painted in accordance with MIL-E-917, except that only one coat of gray enamel shall be applied. The outline and mounting dimensions shall be as specified on figure 2. Touching up shall be permitted for marks or scratches due to assembly, testing or other factory handling. The metering system shall conform to MIL-E-2036. Class III metering systems shall have individual cases and panel covers for the three displays, see Figure 3.

3.5.5 Electronic parts. Electronic parts utilized in the metering system shall conform to MIL-STD-2036. Approval of nonstandard parts shall be required prior to submission of first article sample for testing or if first article tests are waived, prior to production of metering systems under the contract.

3.5.6 Edgeboard connectors. The edgeboard connectors shall be in accordance with MIL-C-21097.

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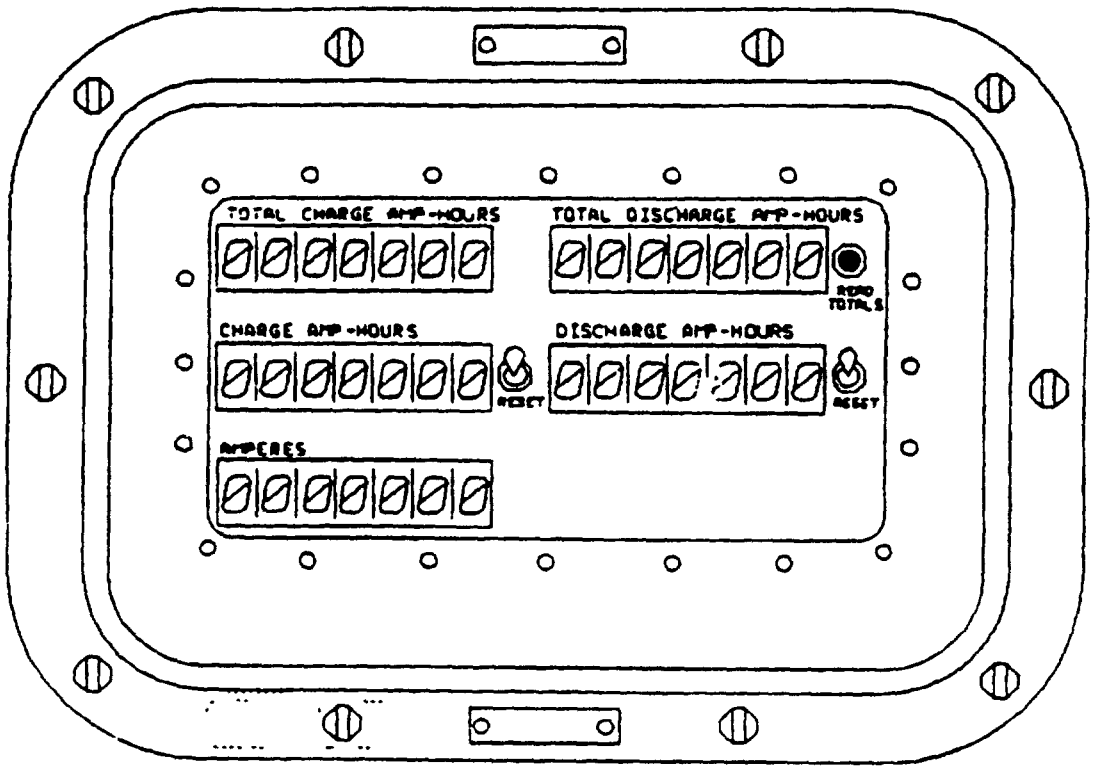
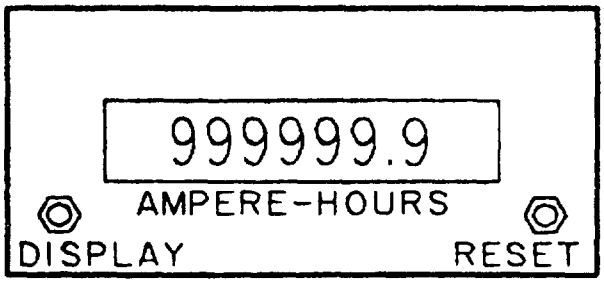
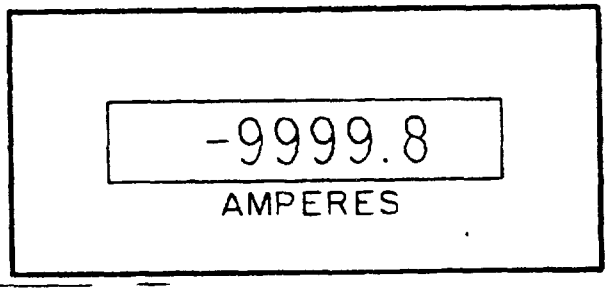


FIGURE 1. Display arrangement, Classes I and II

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Dimensions:
Panel Cover: 48 -0mmH x 96.3mmW x 12.7mmD
Case: 44.96 ± .76H X 92.47 ± .76W x 125.5mmD
(including connections)



Dimensions:
Panel Cover: 72.2mm H x 144mmW x 12.7mmD
Case: 6858 ± .78mm H x 137.67 ± .78mm x 173.00mmD
(including connections and fuseholder)

NOTES:
1. DIMENSIONS ARE IN mm.
2 UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE PLUS OR MINUS ONE mm

FIGURE 2: Outline and mounting dimensions, Classes I and II.

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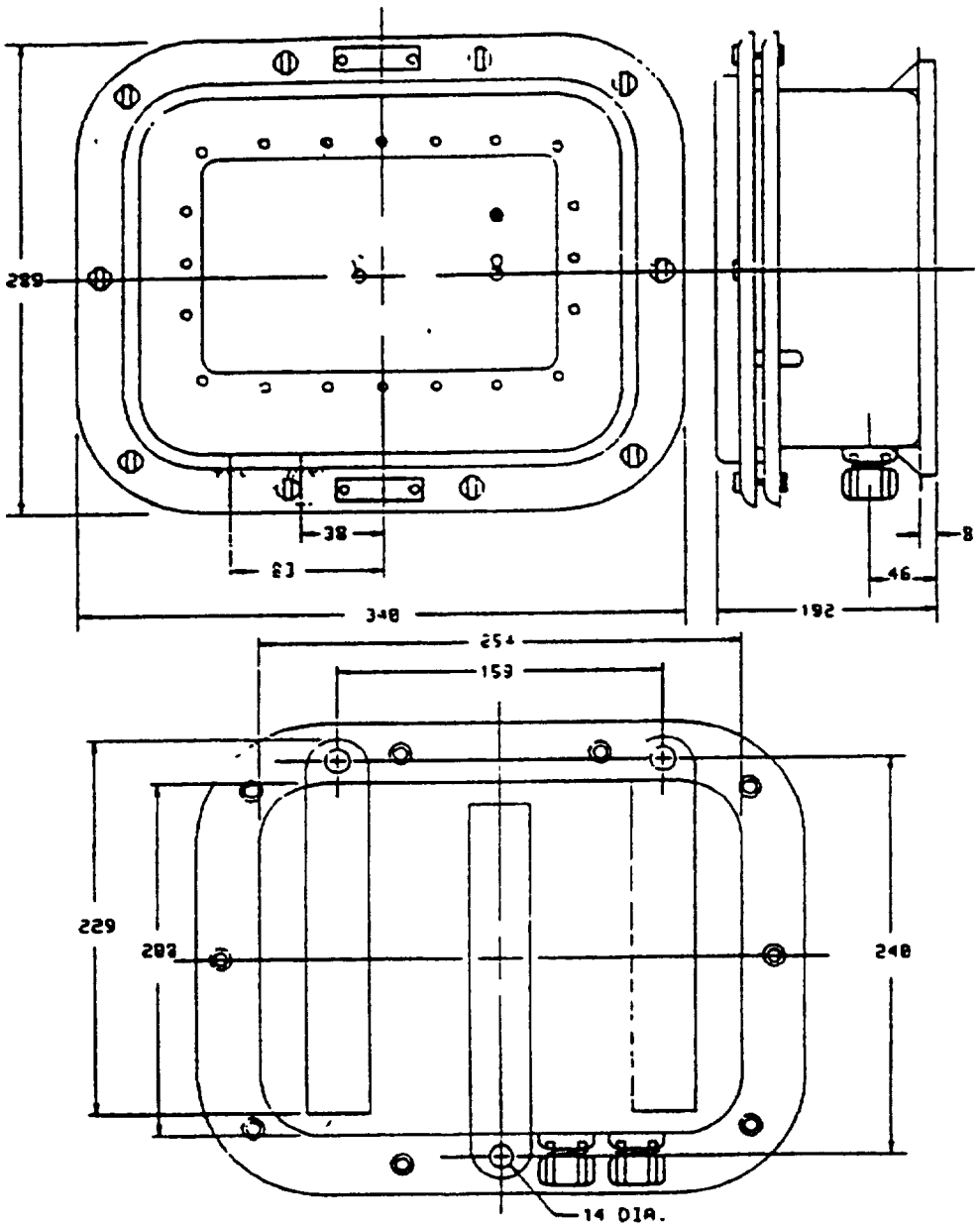


FIGURE 3: Display arrangement, Class III.

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3.6 System connections and isolation. The enclosure shall be provided with stuffing tubes as specified in 3.4.3 and terminal boards as specified in 3.5.2 as required to make electrical connections to the shunt, the alternating current (ac), power source, and the metering systems subsystems. The current shunt shall be connected to the terminal board within the enclosure which is normally inaccessible to the operators; but may be exposed while subsystems are being replaced or adjusted for calibration purposes. Class III systems shall be connected in accordance with the system drawings and technical manual, see 6.3 and 6.4.

3.6.1 Insulation resistance. A minimum insulation resistance of 10 megohms shall be maintained between the direct current (dc) system being metered, the ac power line, and the metering system.

3.6.2 Dielectric stress. A minimum dielectric withstand voltage of 1,500 volts root mean square (Vrms), 60 hertz (Hz), shall be maintained between the dc system being metered, the ac power line, and the metering system subsystems. There shall be no evidence of arcing, corona (visible, audible, or odorous), flashover, or punctured insulation following the dielectric withstand test.

3.7 System performance.

3.7.1 Accuracy. The accuracy of the metering system shall be:

3.7.1.1 Conditions. The reference conditions for the metering system accuracy shall be:

a. Current: zero to 10,000 amperes (A)
b. Temperature: 20 to 30 °C
c. Relative humidity: 10 to 60 percent
d. Power line voltage: 110 to 120 Vac
e. Power line frequency: 57 to 63 Hz
f. Operating position. any
g. Warm-up-time: not more than 5 minutes.

3.7.1.2 Requirements.

a. For current (when displayed) - plus or minus 0.3 percent of reading plus 0.2 amperes (A)

b. For AMP HRs - plus or minus 0.3 percent of reading plus 0.2 times the integration time plus one least significant digit.

For errors due to environmental conditions (see 3.10).

3.7.2 Input power: Instruments of the metering system shall be connected to and capable of operating from electrical power having characteristics as specified for 115 Vac, single phase, type I of MIL-STD-1399, section 300. The display indications shall not be affected to an extent greater than that specified in 3.10. System calibration shall not be permanently affected by voltage or frequency excursions specified in 3.10 The power required from the 60 Hz source to operate the system shall be not greater than 25 watts (W) under the reference conditions of 3.7.1

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3.7.3 Excessive dc signal input. Although over-range point for the metering system shall be at least 10,000 A, no damage to the system modules or subsystems or permanent effect upon system calibration shall result from fault currents to 65,000 A of any duration or from transient changes in busbar currents at any rate. This tolerance does not necessarily apply to the current measuring shunt, which is not a supplied part of the system.

3.7.4 Busbar voltage. The nominal busbar voltage is 270 volts direct current (Vdc), with a range of 175 Vdc to 355 Vdc with one battery of 126 or 128 cells.

3.7.5 Calibration. All calibration adjustments for the system shall be located within the system enclosure. Calibration shall be unaffected by variations in the lengths or resistance of cables between the system instruments or between the system and the current-measuring shunt so long as these values remain within the limits specified in 3.4. Calibration adjustments shall be unaffected by shock, vibration, or other environmental conditions within the limits specified herein.

3.8 Environmental requirements.

3.8.1 Mechanical.

a. **Shock.** For Classes I and II the metering system shall meet the requirements for type A, grade A, hull (bulkhead) mounted, lightweight, class I equipment as specified in MIL-S-901. After the shock test metering system shall meet the requirements of 3.10 (following reset of the display to zero). Following the shock test of (see 4.6.2.4), the system shall be fully operable. Insulation resistance shall be not lower than the value specified in 3.6.1 and dielectric withstand voltage shall be not lower than 65 percent of the values specified in 3 6.2. Class III systems will be required to pass the shock requirements of the panel in which they will be installed.

b. **Vibration.** For Classes I and II the metering system shall withstand the type I vibration requirements of MIL-STD- 167-1 without change in indication of the instruments. After the vibration test the metering system shall meet the requirements of 3.10 (following reset of the meter to zero). Following the vibration test (see 4.6.2.5) the system shall show no signs of breakage or distortion of parts sufficient to impair operation. Insulation resistance shall be not lower than the values specified in 3.6.1. Class III systems will be required to pass the vibration requirements of the panel in which they will be installed.

3.8.2 Temperature range. The system shall operate in the temperature range of 0 to 50 °C. Reference temperature of initial calibration shall be 25 ± 2 °C. Changes in display indications due to ambient temperature excursions within 0 and 50 °C shall not exceed the values specified in 3.10. The system shall survive storage at temperatures in the range minus 20 to plus 70 °C without damage.

3.8.3 Relative humidity. For Classes I and II the system shall survive storage for 168 hours at 31 ± 5°C at 95 percent plus or minus 5 percent relative humidity. Following such storage, insulation resistance shall be not less than 2 megohms and the components shall not be damaged or corroded.

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3.8.4 Watertightness. For Classes I and II the system shall meet the watertight requirements of MIL-E-2036.

3.9 Inclination. The system operation, performance, and accuracy shall not be affected by mounting position or inclination and shall meet the requirements of MIL-STD-2036.

3.10 Errors due to environmental conditions. In addition to the errors permitted in 3.7.1, errors as shown in Table I also shall be permitted under or immediately following exposure to the extremes of temperature or line power. Requirements for the ammeter apply to class II instruments only (see 1.2.1).

TABLE I. Errors due to environmental conditions.

Condition	Display	Reading
Temperature 0 to 20 °C 30 to 50°C	Ammeter	±0.5 percent of reading
	AHM and totalizing AHM HR indicator	±0 5 percent of reading
Input power line voltage or frequency variation greater than 3 percent	Ammeter	±0.5 percent of reading
	AHM and totalizing AHM HR indicator	±0.5 percent of reading
Vibration (see 3 8)	Ammeter	±0.5 percent of reading
	AHM and totalizing AHM HR indicator	±0 5 percent of reading
Shock (see 3.8)	Ammeter	±0.5 percent of reading
	AHM and totalizing AHM HR indicator	±0.5 percent of reading

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3.11 Maintainability. All metering system subsystems shall conform to the design-for-maintainability requirements of MIL-STD-470 and publication NAVSEA 0967-LP-00-312-8010. Positioning and alignment of all subassemblies shall employ positive means to ensure correct reassembly. Connectors for a given subassembly shall not be physically interchangeable unless the subassembly is also interchangeable with regard to function and performance. The maximum equipment repair time (ERT) shall be 1 hour.

3.11.1 Test equipment. The system shall be capable of being maintained, repaired, and calibrated with the aid of only the following test equipment selected from MIL-STD-1364.

a. Multimeter
b. Oscilloscope
c. DC voltage calibrator
d. Clock or watch with seconds indication

3.11.1.1 Unspecified test equipment. A contractor desiring to use test equipment other than that specified herein shall submit the list to the Contracting Officer with justification of the necessity for using such test equipment prior to the start of manufacture of the system.

3.11.2 Repair of removable parts and subassemblies. The repairability of any removable modules, parts, or subassemblies shall be determined in accordance with MIL-E-917. The maximum time required to repair any malfunctioned parts, module, or subassembly, after removal from the system, shall be 10 minutes.

3.12 Reliability. The metering system shall have a minimum mean-time-between-failure (MTBF) of 10,000 hours. Failure is defined as an increase of errors of indication beyond the limits specified (see 3.7.1) or failure to indicate.

3.13 Identification of equipment division by reference designation. To provide identification that will aid documentation and to facilitate maintenance, each part, subassembly or assembly of the current measuring system shall be identified by reference designations in accordance with ANSI 200-75

3.14 Identification plate. The current measuring system shall be equipped with an identification plate in accordance with P-15024. The identification plates shall conform to the marking requirements of P-15024/5.

3.15 Safety. Safety requirements shall be in accordance with MIL-STD-454, requirement 1.

3.16 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. Items shall meet the requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that products or supplies submitted to the Government for acceptance comply with the contract requirements. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table II.

TABLE II. Inspections

Inspection	First article inspection	Quality conformance inspection	Requirement	Test method
General examination	X	X	3.5	4.6.1
Enclosure	X		3.5.4	4.9
Insulation resistance	X	X	3.6.1	4.6.3
Dielectric withstand voltage	X	X	3.6.2	4.6.4
Accuracy	X	X	3.7.1	4.6.2
Input power variation	X	X	3.7.2	4.6.2.2
Sustained high current and short circuit	X		3.7.3	4.6.2.3
Calibration	X	X	3.75	4.6.6
Shock	X		3.8.1(a)	4.6.2.4
Vibration	X		3.8.1(b)	4.6.2.5
Temperature cycling	X		3.8.2	4.6.2.6
Humidity	X		3.8.3	4.6.3.1
Inclination	X		3.9	4.6.2.7
Maintainability demonstration	X		3.11	4.7
Reliability	X		3.12	4.8

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4.4 Quality conformance inspection. Each metering system offered for delivery shall be subjected to the quality conformance inspection as specified in table II (see 6.3).

4.5 Test conditions. Unless otherwise specified in the applicable test paragraph, all tests shall be conducted under the conditions specified in 3.7.1. The specific ambient temperature at the time of test shall be measured and recorded. Adjustment and calibration of the system shall be made prior to starting the tests. No adjustments or calibration shall be allowed during or between the tests.

4.6 Inspection procedure.

4.6.1 Visual examination. Units and assemblies of the metering system shall be visually examined to determine that:

a. Parts used are as identified by drawings

b. Parts are mounted as shown on the drawings. Examination for correct location, orientation, securing method (fasteners, insulators, and others), and torquing where applicable are included

c. Marking of parts (or, where parts are too small to mark, marking of adjacent panel) and terminals is complete and consistent with schematic and connection diagrams

d. Hookup wire complies with the drawings

e. Hookup wire is neatly formed into groups and laced, tied, or clamped securely into place (with nonflammable cord or clamps) to prevent chafing due to vibration

f. Wire ends are marked

g. Stuffing tubes are provided where wires pass through holes in metal panels

h. Wires are not bent sharply around corners or around corners having sharp edges

i. Wires are connected by solder or other mechanical connections (with locking devices)

j. Indicator mounting panel drilling plan and mounting means are as specified

k. Solder joints are proper

4.6.2 Accuracy determination.

4.6.2.1 Accuracy. The metering system shall be operated at conditions as specified in 3.7.1 and readings made at busbar current and AMP HR values of table III. A millivolt source may be used in lieu of current source and shunt.

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TABLE III. Accuracy.

Charge busbar current	Discharge busbar current	Ampere-hours (minimum)
1	1	3
5[1]	5[1]	3[1]
50	50	5
250	250	125
500[1]	500[1]	250[1]
800	800	250
1,000	1,000	50
2,500	2,500	250
5,000	5,000	100
6,000[1]	8,000[1]	100[1]

[1]only these values are required for quality conformance tests, shock test, vibration tests, and temperature cycling.

4.6.2.2 Input power variations. Input voltage and frequency shall be adjusted to the following values to check the system accuracy at 5 A, 500 A, and 800 A of charge and 800 A of discharge busbar current (use table III for the minimum AMP HRs required for each value of current):

a. 103.5 V at 63 Hz
b. 103.5 V at 57 Hz
c. 126.5 V at 63 Hz
d. 126.5 V at 57 Hz

4.6.2.2.1 Errors. If data shows that errors are equal to or less than 2/3 of the sum of the applicable values specified in 3.7.1 and table I, no additional values of busbar current need be checked. Otherwise, each value of table III shall be checked Voltage shall also be adjusted to 103.4 V at 60 Hz, then 126.5 V at 60 Hz for 15 minutes each, then indication at 50 amperes of busbar current checked.

4.6.2.3 Sustained high current and fault current. The accuracy test specified in 4.6.2.1 shall be made after the system has been operated as follows:

a. 1,000 amperes for 10 minutes
b. 8,000 amperes for 10 minutes
c. 65,000 amperes for 1 second*.

*The overload indication on the ammeter display should be monitored during the 65,000 accuracy test (see 3.3 (a)).

4.6.2.4 Shock. The system shall be mounted on a fixture that simulates an actual installation. The assembly shall then be tested as specified in MIL-S-901 Tests shall be performed with approximately 50 A of busbar current flowing

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4.6.2.4.1 Shock test failure definition. Following the shock test specified in 4.6.2.4, failures are as follows:

a. Failure to indicate or errors of indication exceeding the values of table I as determined by the accuracy checks specified in 4.6.2.1

b. Breakage or distortion of any parts sufficient to impair operation

c. Insulation resistance lower than the value specified in 3.6.1

d. Dielectric withstand voltage lower than 65 percent of the specified value specified in 3.6.2.

4.6.2.5 Vibration. Using the same fixture as specified in 4.6.2.4, the system shall be subjected to a type I test of STD-167-1. Failure of this test shall be determined by the same criteria as specified in 4.6.2.4.1, except that dielectric withstand voltage need not be determined.

4.6.2.6 Temperature cycling. The system shall be subjected to temperature cycling over the rage of temperatures specified in 3.8.2. The test shall be performed to check the system accuracy as specified in 4.6.2.1 as follows:

a. Determine accuracy at reference conditions

b. Condition the system at minimum temperature for 8 hours

c. Determine accuracy at the minimum temperature

d. Condition the system at reference temperature 25 ± 2 °C for 8 hours

e. Repeat b, c, and d twice for a total of three cycles to minimum temperature and return to reference temperature

f. Condition the system at maximum temperature (50 °C) for 8 hours

g. Determine accuracy at maximum temperature (50 °C)

h. Condition the system at reference temperature 25 ± 2 °C for 8 hours

i. Repeat f, g, and h twice for a total of three cycles to maximum temperature and return to reference temperature

j. Maximum change in indication between observations at minimum and maximum temperatures shall be considered to be the effect of temperature.

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4.6.2.7 Inclination. Using the fixture as specified in 4.6.2.4, the system shall be 'inclined forward, backward, to the right, and to the left to an angle of 60 degrees from the vertical to test compliance with accuracy requirements of 3.7.1. The accuracy shall be checked at busbar current and AMP HRs reading as follows:

Charge busbar current	Discharge busbar current	Ampere-hours
50A	50A	5 (minimum)

4.6.3 Insulation resistance. Insulation resistance test shall be conducted in accordance with method 302 of STD-202. Test conditions shall be as follows:

a. Test potential: 500 Vdc plus or minus 10 percent
b. Points of measurement: As specified in 3.6.1
c. Electrification time: 60 seconds minimum.

4.6.3.1 Humidity. A system shall be stored for 168 hours at 95 percent ± 5 percent relative humidity at 31 ± 2 °C. Upon completion of this period, the components shall be tested as specified in 4.6.3 for compliance with 3.8.3. The test shall be performed within 5 minutes after completion of the humidity period. Following the test, the components shall be carefully examined for any signs of material damage and corrosion.

4.6.4 Dielectric withstand voltage. Dielectric withstand voltage shall be conducted in accordance with method 301 of MIL-STD-202. Test conditions shall be as follows:

a. Magnitude of test voltages. Magnitude shall be as specified in 3.6.2

b. Nature of test voltage. The test voltage shall approximate a true sine wave of a frequency not less than the rated frequency of the circuit under test

c. Points of measurement. Points of measurement shall be as specified in 3.6.2

d. Duration of test. Duration shall be 60 seconds, minimum. The test voltage shall be removed by adjustment of its value to zero, not by sudden interruption

e. Definition of failure. Any evidence of arcing, corona (visible or odorous), flashover, or punctured insulation shall be interpreted as failure of the test

f. Dielectric withstand voltage test. This test shall be conducted after all other tests have been completed.

4.6.5 Temperature rise. The system shall be operated with meters and indicator or simulated impedances connected and also with a shunt connected The shunt current shall be varied to determine the value (within limits corresponding to busbar currents from 0 to 8,000 A) resulting in maximum power dissipation in the ammeter and AHM circuits. The ambient temperature shall be

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maintained at 50 °C. Measurement of temperatures of circuit parts shall be performed as specified in E-917. The power consumed by the control unit shall be measured to determine compliance with 3.7.2.

4.6.6 Calibration demonstration. Resistance shall be inserted between the ammeter and the shunt to simulate various cable lengths. The resistance shall be adjusted to zero and 3.5 ohms per conductor ad compliance with 3.7.5 shall be demonstrated.

4.7 Maintainability demonstration test. The time required to accomplish normal corrective maintenance and preventive maintenance tasks shall be demonstrated (see 6.3 and 6.7). The procedure for the maintainability demonstration shall be the same as the procedures that will be in the equipment technical manual.

4.8 Reliability. Reliability tests shall be in accordance with MIL-STD-781, test plan III, test level B.

4.9 Enclosures. The enclosures shall be inspected to determine conformance with specifications (see 3.5.1 and 3.5.2).

4.10 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packing requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.8.)

5.1 Packing requirements. The packaging (preservation, packing and marking) requirements shall be in accordance with MIL-E-17555 for the level of preservation, the level of packing, including marking and packaging acquisitioning therein, as specified (see 6.2.1). In addition, for Navy acquisitions the following applies:

a. Navy fire-retardant requirements.

1) Lumber and Plywood. Unless otherwise specified (see 6.2.1) all lumber and plywood including laminated veneer materials used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B	Type II - weather resistant Category I - general use
Level C	Type I - non-weather resistant Category I - general use

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2) Fiberboard. Fiberboard used in the construction of interior (unit and intermediate) and exterior boxes including interior packaging forms shall conform to the class-domestic/fire retardant or class-weather resistant/fire retardant material requirements of PPP-F-320.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification is intended to cover a digital current measuring system used to measure and display the instantaneous value and time integral of direct current in a storage battery circuit for submarines and surface ships. The ammeter continuously measures and displays the current into or out of the battery. The AHM continuously records the AMP HRs entering or leaving the battery; therefore, it can be used to monitor the battery state of charge. The totalizing AHM provides a summation of AMP HRs used over the life of the battery.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Class and type (see 1.2.1 and 1.2.2)

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2 1.1 and 2.2)

d. When first article inspection is required (see 3.1)

e When cable is required (see 3.4)

f. When maintainability demonstration test is required (see 4.7)

g. Level of preservation, level of packing and other packaging options required (see 5.1)

h. When fire retardant lumber and plywood is not required (see 5.1 a (1)).

i. Class of fire retardant fiberboard required (see 5.1 a (2)).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
appendix	DI-DRPR-80651	Engineering Drawings Level 3	DOD-E-1000
4.1.1	DI-R-4803	Inspection System Program Plan	-
4.4	UDI-T-23732	Test Procedures	-
4.4	DI-NDTT-80809	Test, report	-
4.7	DI-R-5192	Maintainability Demonstration Report	-

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.5.1 Spare/repair parts. When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a reproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.7 Maintainability demonstration test. The elapsed time and manhours required to replace each of the onboard repair parts should be reported. The elapsed time and manhours to perform each proposed preventive maintenance task should also be reported. These time recordings may be obtained where possible from the maintenance actions associated with other tests required by this specification in order to minimize the amount of special effort required to obtain the maintainability data The time data should include the fault correction time for corrective maintenance tasks and servicing time for preventive maintenance exclusive of any subsequent test time. A maintainability demonstration report should be submitted which contains the data recorded, analysis of the data, and the results of any retests if applicable

6.8 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Subject term (key words) listing.

Ampere-hour
Battery
Digital
Meter

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6625-N969)

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APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included on the drawings when specified in the contract or order. This appendix is mandatory only when data item description DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENTS

30.1 Drawings and certification data. Drawings shall be not smaller than 430 by 560 mm nor larger than 710 by 1020 mm. The size drawing scale shall be so that the number of sheets required shall be the minimum consistent with content and clarity. Drawings shall contain the following data:

a. Manufacturer's name and model number

b. Reference to this specification

c. List of parts, tabulating part identification, including original manufacturer's part number, specification data of part, find number corresponding to part flags of outline drawings

d. Identification and stress level data (part manufacturer's and in-use ratings) of each electrical part

e. Schematic diagrams

f. Shock classification and approval reference

g Outline drawing of each assembly including:

(1) Overall and mounting dimensions

(2) Location and size of mounting holes

(3) Mass of each assembly

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(4) Panel cutout data for the indicator

(5) Position, size and marking of terminals used for power input and system part interconnection

h. Interconnection cable requirements such as permissible resistance, and shielding requirements

i. Amperes and power required from the 115 V, 60 Hz source

j. Graphic or tabular presentation of characteristics of circuits, meters, and indicator. This data is required for trouble-shooting assistance and contain complete calibration procedures and instructions. Normal readings at relevant circuit points under specified condition shall be presented.

k. List of recommended test equipment such as multimeters or portable standard instruments

l. Schematic circuit of the complete system

m. Connection diagram showing approximate part location, cable routing within equipment, and terminal and wire marking

n. Description of operation, describing briefly how the circuit parts perform their functions to yield specified systems operation.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-24614A(SH)	12 FEBRUARY 1993

3. DOCUMENT TITLE
METERING SYSTEM, STORAGE BATTERY CURRENT AND AMPERE-HOUR, DIGITAL (METRIC, NONMERCURY TYPE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER NAVAL SEA SYSTEMS COMMAND SEA 05Q42	(1) Commercial (703) 602-6020	(2) AUTOVON (AV) 332-6020
c. ADDRESS (Include Zip Code) 2531 NATIONAL CENTER BLDG 3 WASHINGTON, DC 20362-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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